

myWealthPilot

Infrastructure & Platform Overview

Generated: May 11, 2026 | All services on Free Tier

1. Web Hosting - Render.com

Platform	Render.com (Free Plan)
CPU	Shared CPU (0.1 vCPU)
RAM	512 MB (hard limit, SIGKILL on exceed)
Workers	2 Gunicorn sync workers, 120s timeout
Bandwidth	100 GB/month
Auto-deploy	Yes - on git push to main branch
Sleep Behavior	Spins down after 15 min idle
Keep-Alive	cron-job.org pings /login every 14 min
Cold Start	~30-50 seconds after idle sleep
SSL	Auto (Let's Encrypt)
Region	Oregon (US West)
Health Check	/health endpoint
Runtime	Python 3.12.0

2. Database - Neon.tech PostgreSQL

Platform	Neon.tech (Free Plan, no expiry)
Engine	PostgreSQL 17
Storage	0.5 GB limit (~34 MB used, 6.77%)
Tables	24 tables
Records	61+ rows
Compute	Autoscaling up to 2 Compute Units
Scale-to-Zero	Yes - scales down when idle
Branches	Up to 10 per project
Region	US East 1 (N. Virginia, AWS)
Connection	SSL required (sslmode=require)
Previous DB	Render PostgreSQL (migrated, expires May 12, 2026)

3. Domain & DNS

Domain	mywealthpilot.in
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Render Subdomain	wealthpilot-wcm7.onrender.com
SSL Certificate	Auto-managed by Render (Let's Encrypt)
DNS Target	Render Web Service
Cost	~Rs.500-800/year (domain renewal only)

4. Source Code & CI/CD

Repository	GitHub - vinothkumarmari/WealthPilot
Plan	GitHub Free
Branch	main (auto-deploys to Render)
Build Command	pip install -r requirements.txt && flask db upgrade
Start Command	gunicorn run:application --bind 0.0.0.0:\$PORT --workers 2 --timeout 120

5. Email / SMTP

Provider	Gmail SMTP (smtp.gmail.com:587)
Status	BLOCKED - Render Free blocks outbound ports 25/465/587
Workaround	Admin & trusted users skip OTP verification
OTP Toggle	Global enable/disable + per-user skip from admin panel

6. Payment Gateway

Provider	Razorpay
Plan	Standard (pay-per-transaction)
Integration	Server-side webhook verification

7. Application Stack

Component	Technology
Backend	Flask 3.0, Python 3.12
ORM	SQLAlchemy + Flask-Migrate (Alembic)
Authentication	Flask-Login + OTP (SHA-256 hashed)
Rate Limiting	Flask-Limiter
CSRF Protection	Flask-WTF
Frontend CSS	Bootstrap 5.3.2
Charts	Chart.js 4.4.1
Icons	Material Icons Outlined
Fonts	Google Fonts - Inter
Theme Color	#6C5CE7 (purple gradient)

PWA	Service Worker + manifest.json
Voice Input	Web Speech API (free, no API key)
Offline Storage	IndexedDB for local data sync
Languages	4 - English, Tamil, Hindi, Telugu (850+ keys)
PDF Processing	pdfplumber + Pillow
Dependencies	17 Python packages

8. Limits & Constraints

Resource	Limit
RAM	512 MB (hard limit)
DB Storage	0.5 GB (Neon free tier)
Cold Start	~30-50 sec after idle
SMTP	Blocked (all outbound mail ports)
Concurrent Requests	2 (sync workers)
Load Balancer	None (single instance)
CPU Cores	Shared (no dedicated cores)
Horizontal Scaling	Not available on Free tier
CDN	None
Bandwidth	100 GB/month

9. Monthly Cost Summary

Service	Plan	Cost
Render Web Service	Free	\$0
Neon PostgreSQL	Free	\$0
GitHub Repository	Free	\$0
cron-job.org	Free	\$0
Gmail SMTP	Free (blocked)	\$0
Razorpay	Pay-per-txn	~2% per txn
Domain (annual)	mywealthpilot.in	~Rs.500-800/yr

Total Monthly Cost: Rs.0/month (only domain renewal annually)

10. Architecture Flow

User (Browser/PWA)
|
v

```
mywealthpilot.in (Custom Domain)
  |
  v
Render.com (Gunicorn + Flask, 2 workers, 512MB RAM)
  |
  v
Neon.tech (PostgreSQL 17, 0.5GB, US East 1)

[cron-job.org] --ping--> [Render /login] (every 14 min)
[GitHub main]  --push--> [Render auto-deploy]
[Razorpay]     <-webhook-> [Flask /webhook]
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Document generated for internal reference - myWealthPilot Project